

The Influence of Greening the Suppliers and Green Innovation on Environmental Performance and Competitive Advantage in Taiwan

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Abstract

Recently, many companies have recognized the concepts of green supply chain management or supply chain environmental management. However, relatively little research attention has been devoted to the consideration of relations between greening the supply chain, green innovation, environmental performance and competitive advantage. Hence, this paper aims to bridge this gap by providing empirical evidence to encourage companies to implement green supply chain and green innovation in order to improve their environmental performance, and to enhance their competitive advantage in the global market. A model is constructed to link the aforementioned constructs. Data were collected through a questionnaire-based survey across 124 companies from eight industry sectors in Taiwan. The data are analyzed using structural equation modeling and the results from the final measurement model are used to evaluate the structural model that verifies the significance of the proposed relationships. A prominent result of this study is that greening the supplier through green innovation contributes significant benefits to the environmental performance and competitive advantage of the firm.

Key word: Green Supply Chain Management, Green Innovation, Competitiveness, Environmental Performance.

1. BACKGROUND AND RESEARCH MOTIVATION

With the increasing concern about environmental issues from customers, the public and governments around the world, companies have been developing a number of environmentally friendly programs and “green” products (e.g. green brands and green technologies) ([Hoffmann, 2007](#); [Zhu et al., 2008](#)). It is becoming increasingly important for companies to raise their environmental awareness because more and more international customers and buyers are now requiring their suppliers to produce products that do not contain hazardous and toxic substances. They are also increasingly expecting their suppliers to reduce their consumption of natural energy during the production process in order to reduce the negative impacts of production on the environment. According to Green et al., (1995), the main driving force to implement environmental management is compliance with regulation and legislation and cost saving. [Bowen et al. \(2001\)](#) also indicated that companies will implement green supply chain management (GSCM) practices if they can gain both financial and operational benefits. Well-designed environmental standards can increase a producer’s initiatives to innovate green products and green technologies. Shrivastava (1995) suggests that companies can differentiate their products, improve product quality and lower the cost of production through product and process innovations. They can also extend the environmental concept into their packaging design considerations.

Continuous innovation is a vital solution to overcome pressures from customers, competitors, and regulators (Porter and Van der Linde, 1995). Therefore, it is very important to adopt green innovation and implement Green Supply Chain Management (GSCM) practices within an organization’s value chain (Steger, 1993). Rao & Holt (2005) conducted empirical research on GSCM practices and their relationship between competitiveness and economic performance. Their research showed that implementing GSCM can improve a company’s competitiveness and economic performance. This is in line with previous research which has found that organizations can further reduce production costs and increase economic efficiency by applying environmental related initiatives (Porter, 1991). Improvements in corporate environmental performance and compliance with environmental regulations can also contribute to a company’s competitiveness ([Bacallan, 2000](#)). In addition, the implementation of GSCM through internal and external environmental management contributes substantial benefits by enhancing corporate competitiveness and improving environmental performance ([Rao, 2002](#); [Tukker et al., 2001](#); [Cairncross, 1992](#); [Hart, 1995](#); [Schmidheiny, 1992](#); [Shrivastava, 1995](#); [Porter and van der Linde, 1995](#); [Vermulen, 2002](#)).

Green innovation can be classified into three main categories: green product innovation; green process innovation; and green managerial innovation (Chen *et al.*, 2006; Chen, 2008). Chen et al (2006) found that green product and green manufacturing process innovations are

positively associated with corporate competitive advantage. Chen (2008) then introduced the concept of core competencies. His study showed that green core competencies, defined as “the collective learning and capabilities about green innovation and environmental management” (p533), have a positive influence on a firm’s ability to develop green product and process innovations. In turn, this helps to improve a company’s green image and thus competitive advantage. These studies do not consider the impact of green innovation specifically on environmental performance nor the influence of a greener supply chain on green innovation or performance.

Although the green supply chain management literature is rich, relatively little research attention has been focused on Taiwanese industry. For green innovation, Chen et al. (2006) and Chen (2008) are the only studies in this area that focus on Taiwan. In Taiwan, the economy has grown drastically in the past few decades, but this has had a negative impact on the natural environment. Taiwan’s industries are mostly involved in Original Equipment Manufacturing (OEM) and Original Design Manufacturing (ODM). These companies play important roles in global markets as they are the largest partners of the world leading IT organizations (Yu and Hsu, 2002). Globally, Taiwan is a major international producer of information products ([Chen, 2004](#)). The consumption of resources by Taiwanese industry cannot therefore be underestimated.

This research will therefore focus on the Taiwanese context, as it is relatively under-researched. Although other studies have considered the relationship between green supply chain management and performance or competitive advantage, these have not considered the links between a greener supply chain (greening the supplier) and green innovation, or the ability to develop green products and manufacturing processes and green managerial innovation. In this study, these relationships will be examined and subsequently the impact of these factors on both the environmental performance and competitive advantage of organizations. Theoretically, the contribution of this research is therefore to consider the interplay and relationships between all of these factors, which has not previously been considered. Specifically, this study will answer two research questions: (1) To what extent are firms greening their supply chain and implementing green innovation? (2) Are there any significant direct and indirect links between greening the supplier, green innovation, environmental performance and competitive advantage? To answer these research questions, a questionnaire survey was conducted with Taiwanese industry respondents. This paper will present the results of the survey and identify implications for practice and further research needed in this area. Section 2 presents a literature review and develops the hypotheses which form the basis of this study. Section 3 presents the research method and how the data were collected and analysed. Section 4 discusses the key statistical results and Section 5

discusses these results and concludes this paper.

2. REVIEW OF LITERATURE AND HYPOTHESES DEVELOPMENT

Green Supply Chain Management (GSCM) plays an important role in traditional Supply Chain Management (SCM) ([Linton et al. 2007](#)). In the past, a company could gain competitive advantage through implementing SCM practices, such as improving sourcing decisions ([Wouters et al., 2009](#)). However, companies nowadays have to integrate their supply chain with environmental management due to pressures from customers who have increasing environmental concerns. In addition, organizations can generate more business opportunities than their competitors if they can address environmental issues (e.g. by implementing a proper environmental management system) successfully ([Hansman and Claudia, 2001](#)).

Although there are many competing definitions for GSCM ([Zhu and Sarkis, 2004](#)), GSCM can be broadly classified into internal and external environmental management ([Rao, 2002](#)). Internal environmental management focuses on internal support and commitment for GSCM, compliance with certification and the existence of environmental management systems within the organization ([Zhu and Sarkis, 2004](#)). External environmental management is related to greening the suppliers so that they are involved with the organization towards achieving environmental objectives ([Bowen et al., 2001](#); [Lamming et al., 1999](#); [Lloyd, 1994](#); [Rao, 2002](#); [Hamner, 2006](#); [Makower, 1994](#); [Green et al, 1998](#); [Rajagopal and Bernard, 1993](#)). It also includes green purchasing, cooperation with customers, environmental requirements, investment recovery, and eco-design practices ([Rao and Holt, 2005](#)) or green product innovation ([Chen et al, 2006](#)).

In the literature, some studies have provided evidence that integrating suppliers more strongly into the product innovation process can improve a firm's overall performance (e.g. [Bonaccorsi and Lipparini, 1994](#); [Ragatz et al., 2002](#); [Johnsen, 2009](#); [Lau et al., 2010](#)). However, conflicting evidence has been found in other studies, where they show that such integration could result in poor performance; for example, increased development time and development costs ([Zirger and Hartley, 1994](#); [Ittner and Larcker, 1997](#); [Ragatz et al., 2002](#); [Daz et al., 2006](#)). Extending this into the debate around environmental practices, it is therefore important to understand the impact and relationship of more closely integrating GSCM practices and in particular greening the suppliers with green innovation processes to better understand the relationship and impact on both environmental performance and competitive advantage.

2.1. The relationship between greening the supplier and green innovation

Environmental issues are becoming part of strategic planning in organizations due to stricter environmental regulations and customers' increasing concern about environmental issues ([Walton et al., 1998](#)). As a result, long-term strategic advantages can be developed by working closely with suppliers ([Rajagopal and Bernard, 1993](#)). A partnership working and appraisal system is necessary to ensure the appropriate level of product and service quality can be achieved ([Sroufe, 2006](#)). This involves a significant change in the attitudes of those companies who want to establish closer supplier relationships, which requires an investment of time and resources for both parties ([Lettice et al., 2010](#)). Those companies need to work closely with their suppliers to provide sufficient guidance, advice and assistance, and to share their knowledge and skills to help them to become more 'green'. Working towards this, many large companies have established their own environmental standards for their suppliers ([Rao, 2002](#)).

Green innovation is often classified into green product and green process innovations (Klassen & Whybank, 1999; Porter and Van der Linde, 1995; [Hart, 1995](#); [Schmidheiny, 1992](#); Chen et al. 2006; Chen 2008). For example, Klassen and Whybank (1999) categorized green innovation into the product design aspect and the manufacturing process aspect. From the product design aspect, green innovation includes the process to modify an existing product design to reduce any negative impact on the environment during any stage of a product's life cycle. From the process aspect, green innovation is any adaptation to the manufacturing process that reduces the negative impact on the environment during material acquisition, production, and delivery. Green managerial innovation cannot be underestimated in the implementation of GSCM practices. Senior management support is one of the key drivers for the successful implementation of innovation (Hamel and Prahalad, 1989). [Zhu & Sarkis \(2004\)](#), in their research on Chinese manufacturing industry, found that getting commitment from top or middle level managers has a significant influence on the implementation of successful internal environmental management. In this research, we therefore define green innovation as comprising of green product innovation, green process innovation and green managerial innovation.

With increasing concern for environmental issues and regulations, customers and buyers require their suppliers to offer products and materials which take environmental issues into consideration. Therefore, it is necessary to work closely with suppliers early in the product development process to reduce material and packaging which could cause a negative impact on the environment. It is also beneficial to invite suppliers to help to improve the product design and manufacturing processes and to improve overall compliance with environmental regulations. [Rao \(2002\)](#) identified that greening the supplier initiatives do lead to greener

suppliers and more green innovations. However, other studies have found that such integration could result in poor performance; for example, by increasing development time and development costs (Zirger and Hartley, 1994; Ittner and Larcker, 1997; Ragatz *et al.*, 2002; Daz *et al.*, 2006). Finally, studies have shown that internal managerial support for green initiatives is one of the key drivers to successfully implement environmental management systems and practices (Zhu & Sarkis, 2004; Hamel & Prahalad, 1989). To determine whether there is a positive relationship between greening the supplier and green innovation we propose our first hypothesis. Other studies have tended to focus on just green product and process innovation. For this study we separate green innovation into three parts to understand whether greening the supplier influences green product innovation, green process innovation, green managerial innovation, or all three:

H1: Greening the supplier is positively associated with green innovation

- *H1a: Greening the supplier is positively associated with green product innovation.*
- *H1b: Greening the supplier is positively associated with green process innovation.*
- *H1c: Greening the supplier is positively associated with green managerial innovation.*

2.2 The relationship between green innovation and environmental performance

As discussed above, green innovation is composed of green product innovation, green process innovation and green managerial innovation. Chen *et al.* (2006) and Chen (2008) considered how green product and process innovation affect competitive advantage and the green image of an organization, but did not consider how they affected an organization's environmental performance. They also did not consider how green managerial innovation affects environmental performance. Although other studies have considered how GCSM initiatives and greening the supplier impact environmental performance (Porter and Van Der Linde, 1995; Cairncross, 1992; Hart, 1995; Schmidheiny, 1992; Zhu *et al.*, 2008), they have not explicitly considered the links between green innovation and environmental performance. In this study, we therefore propose to contribute to the literature by exploring whether the three categories of green innovation do affect environmental performance. The following hypothesis and sub-hypotheses are therefore proposed:

H2: Green innovation is positively associated with environmental performance

- *H2a: Green product innovation is positively associated with environmental performance.*
- *H2b: Green process innovation is positively associated with environmental performance.*
- *H2c: Green managerial innovation is positively associated with environmental performance.*

2.3 The relationship between green innovation and competitive advantage

Innovation is one of the key forces to increase corporate competitive advantage (Porter and Van Der Linde, 1995). With more environmental regulations, it is becoming increasingly necessary to consider the whole product life cycle when making product and design process decisions. Green products and processes not only reduce the negative impact on the environment, but they can also increase company's competitive advantage (Porter and Van Der Linde, 1995). [Noci and Verganti \(1999\)](#) have investigated this through a qualitative case study and [Chen et al. \(2006\)](#) initiated a survey in the information and electronics industry to consider how green product and process innovations affect competitive advantage. However, there is still a need to provide more empirical evidence across different industries and to consider all three types of green innovation. The following hypotheses are therefore proposed:

H3: Green innovation is positively associated with competitive advantage

- *H3a: Green product innovation is positively associated with competitive advantage.*
- *H3b: Green process innovation is positively associated with competitive advantage.*
- *H3c: Green managerial innovation is positively associated with competitive advantage.*

2.4 The relationship between environmental performance and competitive advantage

In general, companies can improve their environmental performance by implementing environmental management or GSCM. Organizations can add value to core business programs by implementing environmental programs successfully (Hansman and Claudia, 2001). In addition, GSCM promotes efficiency and synergy which includes enhancing environmental performance, improving competitiveness and cost saving (Rao, 2005). Companies can also enhance their green corporate image by improving their environmental performance, which in turn helps to open up new business opportunities and improve competitive advantage (Chen, 2008). [Vachon and Klassen \(2008\)](#) showed a correlation between environmental performance and competitive advantage in their survey of North American organizations. However, Rao's (2002) study of firms in South East Asia (Malaysia, Philippines, Indonesia, Thailand, and Singapore) could not find a correlation between environmental performance and economic performance/competitive advantage. This study will therefore extend this previous research into the Taiwanese context and collect more empirical evidence to see if there is indeed a link between environmental performance and competitive advantage. Therefore, the following hypothesis is proposed:

H4: Environmental performance is positively associated with competitive advantage.

2.5 The relationship between greening the supplier and environmental performance and between greening the supplier and competitive advantage

Working with suppliers on long term strategy to reduce the negative impact on the environment improves corporate competitive advantage. Rao (2002) identified that companies would like to 'green' their supply chains by implementing mentoring methods, and by building stronger relationships and networks. The study also found that companies can improve their environmental performance by working more closely with their suppliers. Similarly, Hamner (2006) found that buyers can improve their environmental performance by educating their suppliers about environmental issues and helping their suppliers to implement environmental management systems. Some studies have attempted to develop supplier selection models from an environmental performance point of view (e.g. [Lee et al., 2009](#); [Bai and Sarkis, 2010](#)). However, they overlooked the effect of greening the suppliers on environmental performance. Although some studies have already identified the importance of some supply chain activities ([Zhu and Sarkis, 2004](#)), more understanding of the specific relationship between greening the suppliers and environmental performance is sought in this study. This leads to the fifth hypothesis:

H5: Greening the supplier is positively associated with environmental performance

Working closely with suppliers can enhance competitiveness because suppliers can provide organizations with useful market information. For example, some of the leading organizations in Taiwan have enhanced their competitiveness in the global market, improved their product quality and reduced production cost through implementing partnering systems ([Rao, 2002](#)). In the past few years, one key success factor for GSCM has been to work with suppliers who have met environmental criteria to reduce the cost of compliance with regulations ([Sarkis, 1999](#)). Many large and medium sized enterprises have worked with their suppliers not only to meet the environmental regulations, but also to organize other environmental activities for their suppliers, such as providing environmental technical advice and awareness seminars to them and inviting suppliers to be involved earlier in the product development process ([Handheld & Nichols, 1999](#); [Rao, 2002](#); [Rao and Holt, 2005](#)). This has led to improved product quality, reduced costs (unnecessary expense) and enhanced competitiveness in the market. Again, this study focuses specifically on greening the supplier, which has received less attention in previous studies and the following hypothesis is therefore proposed:

H6: Greening the supplier is positively associated with competitive advantage

2.6 The relationship between green product, process and managerial innovation

Since in this study we have included three types of green innovation, it is also interesting to

investigate the synergy and correlation among these three variables. Therefore, the following hypotheses are established:

H7: Green product innovation, green process innovation and green managerial are positively associated with each other

- *H7a: Green product innovation is positively associated with green process innovation.*
- *H7b: Green process innovation is positively associated with green managerial innovation.*
- *H7c: Green managerial innovation is positively associated with green product innovation.*

In summary, the literature review indicates that greening the supplier has a positive influence on green innovation (Rao, 2002; Tukker et al, 2001; Porter and Van Der Linde, 1995; Shrivastava, 1995). In addition, green innovation has a positive impact on competitive advantage (Porter and Van Der Linde, 1995; Klassen & Whybank, 1999; [Rao, 2002](#); Rao, 2005; Chen et al., 2006; Chen, 2008). Furthermore, greening the supplier has a positive impact on environmental performance ([Rao, 2002](#); Rao, 2005; Tukker et al, 2001; Crowe and Brennan, 2007) and competitive advantage ([Rao, 2002](#); Rao, 2005). This study aims to formulate the complex relationship between all of the factors: greening the supplier; green product, process and managerial innovation, environmental performance and competitive advantage. These relationships form a conceptual framework, as illustrated in Figure 1, which will be tested in this study. In this study, greening the supplier is an exogenous variable, and green innovation, competitive advantage and environmental performance are endogenous variables. In addition, green innovation is also an intervening variable in this study.

3. RESEARCH METHOD

3.1 Research design

In order to carry out Confirmatory Factor Analysis (CFA) that establishes causal relationships among the constructs shown in Figure 1, an empirical questionnaire survey followed by Structural Equation Modeling (SEM) is employed. SEM is a statistical methodology which provides parameter estimates of the direct, indirect or total direct links and causal relationships between observed or manifest and reference variables and unobserved variables. [Bentler \(1988\)](#) states that SEM theory represents “causal” processes that create observations on multiple variables. SEM can also provide parameter estimates for relationships among unobserved variables (latent constructs) which are measured using indicator variables. In general, an SEM model can be decomposed into two sub-models: a measurement model and a structural model. The measurement model defines relationships between the observed and unobserved variables and evaluates the reliability and validity of the model. In contrast to the

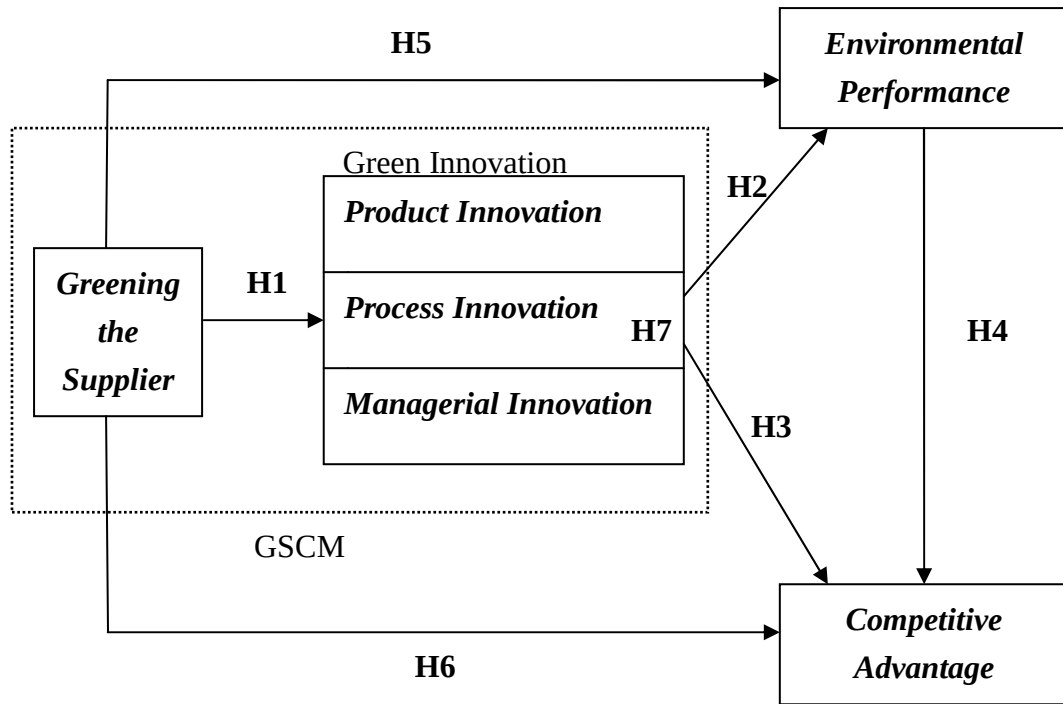


Figure 1: Conceptual Framework

measurement model, the structural model defines the relationships among the unobserved variables and tests the research hypotheses.

For SEM, we have to identify the latent variables that are exogenous and endogenous. Exogenous latent variables are independent variables that are not influenced by other latent variables. Endogenous latent variables are dependent variables that are affected by exogenous variables in the model. In this study, “greening the supplier” is an exogenous variable. “Green innovation”, “environmental performance” and “competitive advantage” are endogenous latent variables. There are different analytical tools to analyze SEM. In this study, AMOS 7.0 is employed. The proposed model (see Figure 1) which would be validated has six latent constructs, each being measured by the indicators in Table 1. A five-point Likert scale (1 = Strongly Disagree and 5 = Strongly Agree) is used as the response format for each indicator variable. The pre-test analysis is illustrated later in this section before analyzing the data for SEM.

Latent construct	Indicator Variables	Code	Mean	Standard Deviation	ITC
Greening the supplier	Selecting suppliers by environmental criteria	Q1011	4.4597	0.75868	0.774
	Requiring and assisting suppliers to obtain a third-party certification of environmental management system (EMS)	Q1021	3.8952	0.75243	0.795
	Providing environmental awareness seminars and training for suppliers	Q1031	3.7581	0.86833	0.719
	Providing environmental technical advice to suppliers and contractors in order to help suppliers to meet environmental criteria.	Q1041	4.1371	0.90468	0.837
	Invite suppliers to join in the development and design stage	Q1051	4.2661	0.85643	0.740
	Sending in-house auditor to appraise environmental performance of supplier.	Q1061	3.8871	1.04539	0.769
Green Product Innovation	Using less or non-polluting/toxic materials. (Using environmentally friendly material)	Q2011	4.2823	0.89770	0.862
	Improving and designing environmentally friendly packaging (eg: less paper and plastic material used) for existing and new products.	Q2021	4.1774	0.83664	0.852
	Recovery of company's end-of-life products and recycling	Q2031	4.2016	0.88327	0.804
	Using eco-labeling	Q2041	4.2984	0.84565	0.879
Green Process Innovation	Low energy consumption such as water, electricity, gas and petrol during production/use/disposal.	Q2051	4.4032	0.83539	0.877
	Recycle, reuse and remanufacture material	Q2061	4.2581	0.89143	0.891
	Use of cleaner technology to make savings and prevent pollution (such as energy, water and waste etc.)	Q2071	4.2984	0.88653	0.904
Green Managerial Innovation	Redefine operation and production processes to ensure internal efficiency that can help to implement GSCM	Q2081	4.3468	0.85582	0.870
	Re-designing and improving product or service to obtain new environmental criteria or directives	Q2091	4.3790	0.83220	0.888
Environmental Performance	Reduction of hazardous waste, emission etc	Q3011	4.2016	0.91046	0.870
	Less consumption of e.g. water, electricity, gas and petrol	Q3021	4.1371	0.88653	0.872
	Improvement of environmental compliance	Q3031	4.1935	0.93424	0.877
Competitive advantage	Customer response	Q4011	4.2903	0.85325	0.886
	Product design and innovation	Q4021	4.2661	0.95516	0.887
	Quality of product and service	Q4031	4.3468	0.87461	0.877
	Low production cost	Q4041	4.2258	0.91810	0.833

Table 1: Indicator variables

3.2 Sample size and data collection

In this study, an online questionnaire survey is the research instrument and an invitation to complete the questionnaire survey was sent out by email. The questionnaire was developed based on the literature presented in preceding sections, and is attached in the appendix of this paper. The questionnaire items regarding the construct “Greening the supplier” (items 1 to 6) are adopted from the studied conducted by Rao (2002) and Rao and Holt (2005). The respondents were asked if the environmental actions had been taken into consideration when selecting and working with suppliers. In addition, the items regarding “Environmental performance” and “Competitive advantage” are also adopted from Rao (2002) and Rao and Holt (2005). They are items 16-18 and items 19-22 within the questionnaire respectively. The other three variables for “Green innovation” are measured based on the items proposed by Chen et al. (2006) and Chen (2008), which can be found in items 7-15 of the questionnaire.

The emailing list had been randomly selected from the online resource “Business Express” (<http://business.com.tw>), a business directory, which lists the firms in Taiwan under different industry sectors. The final sample frame was 1,569 companies from eight industries. Most of Taiwan’s companies do not have a supply chain department to coordinate and manage their upstream and downstream supply chain activities. Therefore, the purchasing department was identified as the target respondent of this study. A covering letter was included to explain the purpose and intention of the survey. It also promised anonymity of both the respondent and the company in the research report.

The invitation was e-mailed directly to the company’s purchasing departments. In order to maximize the response rate, a follow up email was sent to respondents two weeks after sending out the first email. Within two weeks of the first invitation, 70 questionnaires were received. A further 54 questionnaires were received within two weeks of sending out the follow up email. Therefore, a total of 124 completed surveys had been received from 1,569 respondents. All of these returned surveys were complete and acceptable for analysis. Therefore the overall response rate is 7.9%. Based on responses to previous studies (e.g. 72 respondents in Lu’s (2000) study; 83 respondents in Oum and Park’s (2004) study; 72 respondents in Sodhi and Son’s (2009) study), these figures are representative and are of an acceptable standard for publication.

3.3. Pre-test analysis

The main purpose of the pre-test analysis was to validate the sample size and identify whether there was any indication of non-response bias between the two groups of respondents. First of all, the acceptability of the sample size was investigated. Sample size is one of the key issues in SEM because it affects the degrees of freedom of the model and

the model fitting process. In general, when sample size is lower than 100, SEM analysis is not stable (Kline, 1998). According to Bagozzi & Yi (1998), a sample size of between 100 and 200 is required to use the maximum likelihood method. Therefore, the sample size of 124 in this study is acceptable.

The second test was to verify the non-response bias between the two groups of respondents. Initially, 70 questionnaires were received before a follow-up email was sent out, which represents a 4.46% response rate. A second set of 54 questionnaires were received after sending a follow-up email, which represents a 3.44% response rate. Non-response bias was established in two separate tests, the *t*-test and the Chi-Square test.

The *t*-test is the most commonly used statistical method to evaluate the differences in means between two groups. To compare the data from the two different response groups, an independent sample *t*-test was undertaken and the results of the *p*-value for each observed variable was greater than or equal to 0.05, indicating that there was no significant difference between the groups. The data are summarized in Table 2(a). The Chi-Square test was then employed to test the two groups of respondents based on industry sector, ISO14000 certified or not and the number of employees. These results are summarized in Table 2(b). Again, a *p*-value equal to or greater than 0.05 indicated that there was no significant difference between the two groups of respondents. Based on the results of these two tests, there is an indication that there might be no non-response bias.

Table 2. Pre-test for the groups industry, ISO14000 and number of employees

Table 2(a) <i>t</i> -test and <i>p</i> -value	
	<i>p</i> -value
Industry	0.066
ISO14000	0.385
Number of employees	0.355

Table 2(b) Chi-Square and *p*-value

	Chi Square value	Degrees of freedom (df)	<i>p</i> -value
Industry	8.99	7	0.253
ISO14000	0.762	1	0.383
Number of employees	1.648	5	0.895

3.4. Reliability analysis

Reliability analysis is a method to measure the precision or lack of distortion of the indicators. Reliability concerns the consistency, precision and repeatability of the indicator (Kline, 1998). Through the value of Cronbach's Alpha, reliability of each construct and items established in the study are verified. In addition, item-total correlation, composite reliability and variance extracted estimate are used to validate the reliability of each construct through Confirmatory Factor Analysis in the SEM model.

1. Cronbach's Alpha: Ranges between 0.00 (without any reliability) and 1.0 (perfect reliability) inclusive. The larger the Cronbach's Alpha, the better is the consistency in the measurement (Vogt, 1999). Nunnally (1978) suggests that the Cronbach's Alpha should be at least equal to 0.5 and preferably larger than 0.7.
2. Corrected Item-Total Correlation (ITC): Sums up each item correlation and then excludes items with ITC values below 0.4 (Nunnally, 1978), to see if excluding the item increases the Cronbach's Alpha.
3. Composite Reliability: Measures the internal consistency of a single construct. The reliability is higher if the value of composite reliability is greater than 0.6 (Hatcher, 1994). However, a minimum of 0.7 is recommended (Forrel and Larcker, 1981).
4. Average Variance Extracted (AVE): Measures the internal consistency of the model. If the AVE is greater than 0.5, then reliability is good (Hatcher, 1994).

The results of the reliability analysis are discussed in section 4.

4. RESULTS AND ANALYSIS

4.1 Descriptive analysis

The main purpose of using descriptive analysis is to describe the basic features of the data in this study. It provides a simple summary of how the respondents from different industry sectors have implemented greening the supplier and green innovation practices. The size of

the companies in the sample is shown in Table 3.

Table 3. Background of the respondents (N=124)

Number of employees	Frequency	%	ISO14000 (YES)	%	Without ISO14000 (NO)	%
Under 200	29	23.4	7	5.65	22	17.75
200 - 399	14	11.3	7	5.65	7	5.65
400 – 599	7	5.6	6	4.83	1	0.82
600 – 799	8	6.5	8	6.44	0	0
800 - 999	7	5.6	7	5.65	0	0
Over 1000	59	47.6	59	47.58	0	0
Total	124	100	94	75.8	30	24.2

There are 29 companies out of 124 respondents (23.4%) with less than 200 employees, and only 7 of these 29 (24%) are ISO 14000 certified. In addition, there are 95 companies with more than 200 employees (76.6%), and a high percentage (91.58%) of these companies have been ISO 14000 certified. That means the respondents are mostly medium and large enterprises. Table 4 shows the statistics of the respondents with respect to their industry. Most of the respondents (nearly 50%) come from two industries: the computer, electronic product and components industry and the chemical, rubber and plastic product industry. In addition, 24 out of 28 (85%) of the companies in the chemical, rubber and plastic industry have ISO14000 certification. The car and transport equipment sector companies in this sample all have ISO14000 certification.

The mean and standard deviations were also calculated to further explain the current situation for greening the supplier, green innovation, environmental performance and competitive advantage for the Taiwanese firms in the sample. Each of these observed variables were evaluated on a five-point scale (strongly agree, agree, neutral, disagree, and strongly disagree). As can be seen in Table 1, the means for all items across all of the constructs are in the range of 3.76 to 4.46. This implies that many companies in Taiwan have already implemented actions to improve the greening of their suppliers; and to develop green products, processes and managerial innovations and have had corresponding results in terms of good environmental performance and competitive advantage

Industry	Frequency	Percentage (%)	Number with ISO14000 (Yes)	Percentage (%)	Number without ISO14000 (No)	Percentage (%)
Food products and beverages	9	7.3	6	4.85	3	2.42
Computer and electronic products and components	32	25.8	22	17.74	10	8.06
Electric products	15	12.1	9	7.26	6	4.84
Chemical, rubber and plastic products	28	22.6	24	19.35	4	3.23
Machinery and optical equipment	14	11.3	11	8.87	3	2.42
Car and transport equipment	11	8.9	11	8.87	0	0
Service	11	8.9	7	5.65	4	3.23
Others	4	3.2	4	3.23	0	0
Total	124	100	94	75.8	30	24.2

Table 4. Industry of the respondents (N=124)

4.2. Reliability of the measurement model

A total of 22 items (observed variables) are in the model to measure the relationship between greening the supplier, green product innovation, green process innovation, green managerial innovation, environmental performance and competitive advantage. After evaluating the ITC values, it was not necessary to exclude any item in this study because no ITC value was lower than 0.4, as shown in Table 1. The Cronbach's Alpha for six latent constructs were calculated to test the reliability of the questionnaire. The reliability measures are listed in Table 5. The Cronbach's Alpha of each item is more than 0.9 and less than 1.0 ($0.9 < \text{Cronbach Alpha} < 1.0$),

which means that the questionnaire has high reliability and consistency.

Table 5. Descriptive statistics and reliability measures

Latent Construct		Cronbach's Alpha	Composite Reliability	Average Variance Extracted
Greening the Supplier		0.917	0.85	66%
Green Innovation	Product Innovation	0.934	0.77	78%
	Process Innovation	0.936	0.96	88%
	Managerial Innovation	0.92	0.92	86%
Environmental Performance		0.942	0.94	83%
Competitive Advantage		0.954	0.96	86%

In terms of composite reliability, the values for all constructs (0.85, 0.77, 0.96, 0.92, 0.94 and 0.96) are greater than the recommended value of 0.7. That means the measures of the constructs are highly reliable in terms of their internal consistency. In addition, the AVE values for the variables are all greater than the recommended value of 0.5 (66%, 78%, 88%,

86%, 83%, and 86%), which means that those (indicator) variables in the questionnaire have been understood.

4.3. Structural model: statistical significance

Table 6. Support hypotheses: Path coefficient

Hypotheses	Support (Yes/No)	Path Coefficient
H1a: Greening the supplier is positively associated with green product innovation.	Yes	0.99***
H1b: Greening the supplier is positively associated with green process innovation.	Yes	0.93***
H1c: Greening the supplier is positively associated with green managerial innovation.	Yes	0.95***
H2a: Green product innovation is positively associated with environmental performance.	Yes	1.04***
H2b: Green process innovation is positively associated with environmental performance.	Yes	1.26***
H2c: Green managerial innovation is positively associated with environmental performance.	No	-0.13
H3a: Green product innovation is positively associated with competitive advantage.	Yes	0.59***
H3b: Green process innovation is positively associated with competitive advantage.	Yes	0.6***
H3c: Green managerial innovation is positively associated with competitive advantage	Yes	0.72***
H4: Environmental performance is positively associated with competitive advantage	Yes	1.87***
H5: Greening the supplier is positively associated with environmental performance	Yes	0.39***
H6: Greening the supplier is positively associated with competitive advantage	Yes	0.99***
H7a: Green product innovation is positively associated with green process innovation	Yes	0.92***
H7b: Green managerial innovation is positively associated with green process innovation.	Yes	0.94***
H7c: Green managerial innovation is positively associated with green product innovation.	Yes	1.04***

The results from the final measurement model were used to evaluate the structural model that tested the significance of the theorized relationships. The final model with path coefficients is shown in Figure 2. Fourteen of 15 path coefficients are significant. The path coefficients were measured to support hypotheses, as shown in Table 6. In addition, the goodness of fit test is carried out. This test was established by different indices to negate bias related to the use of a single index (Hair et al., 1995 and Bollen and Long, 1993). The sets of goodness of fit indices that were employed in this study are the goodness of fit index (GFI), Chi Square (χ^2 /degrees of freedom (df), the comparative fit index (CFI), the Tucker Lewis index (TLI), the normed fit index (NFI) and the incremental fit index (IFI). According to Bagozzi and Yi (1998), a value of 0.9 is acceptable for the model fit for GFI, CFI, NFI and IFI. [Byrne \(1989\)](#) indicates that the model of fitness (GFI and AGFI) with a range of values between 0.80 and 0.89 is acceptable. Moreover, the modification indices were evaluated in order to modify and increase the validity of the model fitness. Several sets of SEM models and analyses were conducted for the model with alternate path deleted or added to increase the fitness of the model.

Table 7. Measurement of goodness of fit

	Heuristic	Final structural model
Chi-Square	$p > 0.05$	222.6
χ^2 / df	< 3.0	1.66
Goodness of Fit Index (GFI)	> 0.80	0.88
Comparative Fit Index (CFI)	> 0.9	0.98
Normed Fit of Index (NFI)	> 0.9	0.94
Incremental Fit Index (IFI)	> 0.9	0.98
Tucker-Leis Fit Index (TLI)	> 0.9	0.96

The value of the goodness of fit statistics (illustrated in Table 7) were: GFI=0.88 , $\chi^2/df = 1.66$ ($\chi^2=222.6$, $df=134$), CFI=0.98, NFI=0.94, IFI=0.98 and TLI=0.96. Although χ^2 is recognized as the measure of fit, it is influenced by the size of samples (Byrne, 2001). Because of the limitation of sample size, it is replaced by the ratio of χ^2 to the degree of

freedom (χ^2/df), and it is acceptable if the value is less than 3.0 (Carmines and McIver, 1981; Marsh et al., 1998). In this study, the value of χ^2/df is 1.66, which is therefore acceptable. In addition, the GFI is within the acceptable levels of 0.80-0.89, which represents the overall degrees of freedom in the specified model. In contrast to GFI, the next set of goodness of fit statistics was measured for incremental and comparative indexes of fit (Hu & Bentler, 1995). One which should be considered in Baseline Comparisons is CFI, which is recognized as an index to reflect all sample sizes and measure the comparative reduction in noncentrality (Bentler, 1990). The value is 0.98 in this model and is highly acceptable. Moreover, if the value of TLI, which is 0.96 in this study, is larger than 0.95, it is considered a good indicator of good fit (Hu and Bentler, 1999). Next, the values of NFI and IFI are 0.94 and 0.98 respectively, which fall into the acceptable ranges. Bentler (1990) indicates that NFI can estimate the model fitness based on small sample sizes. Based on these values, the final structural model is acceptable, even under the limitation of a small sample size.

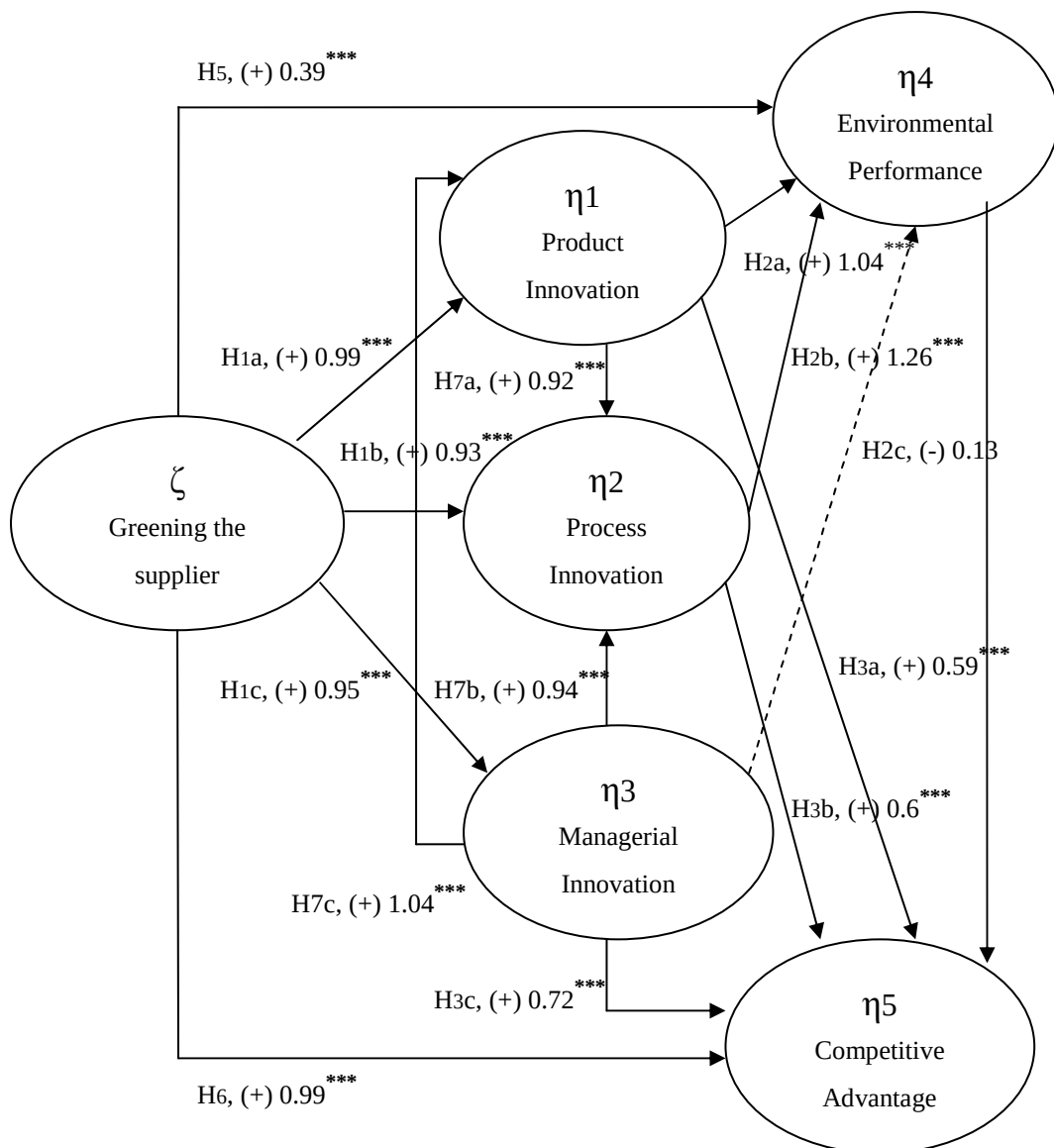
4.4. Results and discussion for structural equation model

The measurement model measures the reliability of each latent construct; and the structural model with path coefficients are to test the 15 hypotheses, as shown in Table 6. The final structural model with path coefficients is illustrated in Figure 2. The path coefficients are measured to test 15 hypotheses between five latent constructs: “Greening the supplier” represents an independent variable (ζ_1); whereas the others are dependent variables, which are “green product innovation” (η_1), “green process innovation” (η_2), “green managerial innovation” (η_3), “environmental performance” (η_4) and “competitive advantage” (η_5). The causal effects between these latent variables are also measured. The final survey data support fourteen of the fifteen hypotheses in this study, as shown in Table 6.

First of all, the association of greening the supplier with environmental performance (H5) is significant ($\gamma=0.39$, $p<0.001$). More importantly, the path association between environmental performance and competitive advantage is strong and highly significant ($\beta = 1.87$, $p<0.001$), which supports H4. Secondly, green process innovation (H2b) and green product innovation (H2a) are positively associated with environmental performance ($\beta = 1.26$ and 1.04 , respectively; $p<0.001$). However, green managerial innovation (H2c), surprisingly, did not show a significant relationship with environmental performance. ($\beta = -0.13$, $p=0.21$). In other words, green product and process innovations may be more effective than green managerial innovations in improving a company’s environmental performance. This might be because managerial innovation has a more indirect effect on the measurement of environmental performance, compared to the other two types of innovation. Managerial innovation relates

to high level strategy and hence it is positively linked to competitive advantage (H3c to be discussed below). Normally, the managerial innovation will be implemented through process

Figure 2. The final structural model with path coefficient between latent constructs



(*, **, *** represent p level of 0.05, 0.01, and 0.001, respectively.)

or product innovation, which can be verified by H7a and H7b, to be discussed later. The results imply that green managerial innovation may have a moderating effect on green product and process innovation, which is further discussed in Section 5.

Thirdly, the associations between greening the supplier and green product innovation (H1a), green process innovation (H1b), and green managerial innovation (H1c) are also highly significant ($\gamma = 0.99, 0.93$, and 0.95 , respectively, $p < 0.001$). Likewise, the path associations between greening the supplier and competitive advantage ($\gamma = 0.99$, $p < 0.001$) provide support for H6. In addition, green product innovation (H3a), green process innovation (H3b) and green managerial innovation (H3c) all have positive path associations with competitive advantage ($\beta = 0.59, 0.6$, and 0.72 , respectively, $p < 0.001$). This implies that a company can gain competitive advantage through the implementation of green product innovation, green process innovation and green managerial innovation.

Finally, the associations between green managerial innovation and green product innovation (H7c) and green process innovation (H7b) are significant ($\beta = 1.04$ and 0.94 , respectively, $p < 0.001$). Similarly, the path association between green product innovation and green process innovation ($\beta = 0.92$, $p < 0.001$) provides support for H7a. Green product innovation therefore contributes to green process innovation, and green managerial innovation contributes to both green product innovation and green process innovation, as shown in Figure 2.

Table 8. Effect of green supplier on competitive advantage

(a) Standardized direct effects

	Greening the Supplier	Green Innovation
Green innovation	2.87	0
Competitive advantage	0.99	1.91

(b) Standardized indirect effects

	Greening the Supplier	Green Innovation
Green innovation	0	0
Competitive advantage	5.48 (2.87*1.91)	1.1

More importantly, the standardized direct and indirect effects between greening the supplier and green innovation are shown in Table 8. The direct effect of greening the supplier to competitive advantage is 0.99, and the indirect effect through green innovation between greening the supplier and competitive advantage is 5.48. The indirect effect (5.48) for greening the supplier to competitive advantage with the intervening variable of green innovation is greater than the direct effect (0.99) for greening the supplier to competitive advantage. Therefore, the intervening variable of green innovation contributes to the relationship between greening the supplier and competitive advantage.

5. DISCUSSION AND CONCLUSIONS

There are some limitations of this survey research. First of all, a total of 124 questionnaires have been collected from 1569 organizations. Although this sample size is acceptable for this study, the response rate is still relatively low, which might affect the representativeness of this sample to all enterprises in Taiwan. In addition, the sample was designed to cover a variety of industry sectors, rather than to focus on a single industry sector. Chen et al, (2006) had only focused on the information and electronics sector and this study aimed to extend their study beyond a single sector in Taiwan. However, as a consequence, the results of our study may not be sufficiently specific for a single industry or sector. Another limitation of this research is related to common method bias. Since the survey was conducted with a single questionnaire (i.e. common source), measurement error may be introduced (Podsakoff *et al.*, 2003). The authors have, however, followed the procedures suggested by Podsakoff *et al.* (2003) and [Menguc and Auh \(2008\)](#) in order to reduce some of the risks for common method bias. They include: (i) protecting respondent anonymity and reducing evaluation apprehension; (ii) careful construction of scale items to avoid item ambiguity; (iii) separating or combining the scale items from different constructs; and (iv) statistical tests to negate bias related to the use of a single index. The final limitation is that the questionnaire survey was not pilot tested prior to the full scale survey. However, in view of the reliability tests presented in this paper, where all Cronbach's Alpha are above the accepted standard 0.7, the survey was shown to be reliable, even though it was not pilot tested (whereas in Chen et al's (2006) survey, some of the Cronbach' Alpha values fell below the 0.7 threshold).

Despite these limitations, this research provides significant insights regarding the deployment of green supply chain management practices (in terms of greening the supplier and green innovation) in Taiwanese industry, and the implications for performance (in terms of environmental performance and competitive advantage). The results in the previous section demonstrate that greening the supplier leads to green product innovation, green process

innovation, and green managerial innovation. The results of this study also show that a company can acquire competitive advantage by engaging in green product innovation, green process innovation and green managerial innovation. In addition, the results of this study show that greening the supplier leads to competitive advantage. More importantly, the findings support that the intervening variables of green product, process and managerial innovation contribute to competitive advantage. More specifically, the indirect effect between green innovation and greening the supplier leads to better competitive advantage than the direct effect between greening the supplier and competitive advantage.

However, the results supported only two of the three hypothesized relationships between the three constructs for green innovation (i.e., green product innovation (H2a), green process innovation (H2b), green managerial innovation (H2c)) and environmental performance. The path coefficient between green managerial innovation and environmental performance is not significant. This implies that green managerial innovation is not as effective as green product and process innovation for improving a company's environmental performance. This is possibly because green managerial innovations are normally implemented through green product and process innovations. This should be further investigated in future research, which should study the moderating effect of green managerial innovation on the relationship between green product and process innovations and environmental performance.

The findings of this study show that Taiwanese companies have started to implement actions towards greening their suppliers and developing greener products and manufacturing processes. The results lead to two key implications for business practice, which are discussed in the next two subsections.

5.1. Green innovation leads to competitive advantage

Environmental management and innovation are predicted to be the most important performance indicators for the competitive advantage of firms in the future. Although green managerial innovation may not necessarily result in improved environmental performance, by focusing on green product and process innovation and green managerial innovation, firms will gain cost savings, increased efficiency, increased productivity and better product quality, all leading to improved competitive advantage. As well as enhancing the green reputation of the firm, there will also be increased opportunities to enter new markets through green product innovation.

One potential benefit of green innovation is that it increases the entry barriers of other competitors. Moreover, reducing pollution and limiting hazardous and toxic waste can improve green efficiency; reduce the cost of hazardous waste disposal; improve compliance

with regulation; improve the response to external environmental pressures from customers, and enhance global competitive advantage with better product quality. Therefore organizations should implement environmental management, and integrate green innovation into business strategies in order to build and maintain competitive advantage.

However, green managerial innovation may not necessarily lead to good environmental performance. Therefore, top management should focus predominantly on green product and process innovations, as there is a stronger link between these and environmental performance.

5.2. Greening the suppliers for a win-win situation

The results of this study show that encouraging green suppliers leads to internal green product, process and managerial innovations, which in turn enhance competitive advantage. Therefore, this research suggests that companies should work closely with, and even integrate their business processes with, their upstream and downstream suppliers to achieve environmental goals and to relieve the pressure on the buyer's requirements. More business opportunities may then be generated by these organizations than by their competitors in the global market.

In practice, companies should devote substantial effort to provide technical support, assistance and guidance to suppliers to implement environmental management systems and to organize environmental awareness seminars or training sessions to help their supplier to improve environmental performance. This will not only improve competitive advantage as discussed before, but also raise suppliers' environmental performance and productivity, which will result in a win-win situation between the two parties.

Although implementing environmental management systems involves capital outlay, it is a valuable investment in the future. Organizations should continuously invest in GSCM, and in particular in greening the supplier and green innovation, to meet tighter and tighter environmental regulations and to build and maintain sustainable competitive advantage in the global market.

In summary, greening the suppliers and green innovation are highly related to the environmental performance of firms and their competitive advantage. Companies should therefore devote substantial efforts to address environmental issues along their supply chains in order to survive and maintain competitive advantage. Future research should be conducted with an even larger scale survey within a single country and across more sectors. It would also be useful to conduct a comparative survey across countries, including both developed and developing countries, with industries and economies at different stages of

maturity. This survey was focused particularly on the links between greening the supplier and green innovation and subsequent environmental and competitive performance. Future studies could extend this to other GCSM practices as well as exploring in more detail the moderating effect of green managerial innovation on green product and process innovations.

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Appendix – The questionnaire

Part 1 Has your company ever taken the following action with your main suppliers or subcontractors?					
SD - Strongly Disagree, D – Disagree, N – Neutral, SA – Strongly Agree, A –Agree					
	SD	D	N	SA	A
1. Selecting suppliers or subcontractors based on environmental criteria	☐	☐	☐	☐	☐
2. Requiring suppliers or subcontractors to obtain a third-party certification of environmental management system (EMS) such as ISO14000	☐	☐	☐	☐	☐
3. Providing environmental awareness seminars and training sessions for suppliers	☐	☐	☐	☐	☐
4. Providing environmental technical advice to suppliers and subcontractors to help them to meet environmental criteria.	☐	☐	☐	☐	☐
5. Inviting suppliers to join in the early product design and development.	☐	☐	☐	☐	☐
6. Sending in-house auditors to appraise the environmental performance of suppliers	☐	☐	☐	☐	☐
Part 2. Has your company ever taken the following action when designing the product?					
SD - Strongly Disagree, D – Disagree, N – Neutral, SA – Strongly Agree, A –Agree					
	SD	D	N	SA	A
7. Using less or non-polluting/toxic materials. (i.e. using environmentally friendly material)	☐	☐	☐	☐	☐
8. Designing or improving environmentally friendly packaging (e.g. use less paper and plastic materials) for existing and new products	☐	☐	☐	☐	☐
9. Recovering company's end-of-life products and recycling	☐	☐	☐	☐	☐
10. Using eco-labeling	☐	☐	☐	☐	☐

Part 3. Has your company ever taken the following action in the production process?					
SD - Strongly Disagree, D – Disagree, N – Neutral, SA – Strongly Agree, A –Agree					
	SD	D	N	SA	A
11. Lower consumption of e.g. water, electricity, gas and petrol during production/use/disposal	☐	☐	☐	☐	☐
12. Recycle, reuse and remanufacture materials or parts	☐	☐	☐	☐	☐
13. Use of cleaner or renewable technology to make savings (such as energy, water, waste.)	☐	☐	☐	☐	☐
14. Redesign of production and operation processes to improve environmental efficiency	☐	☐	☐	☐	☐
15. Redesigning and improving products or services to meet new environmental criteria or directives (such as WEEE/RoHs/REACH/EuP directives etc.)	☐	☐	☐	☐	☐
Part 3. Has your company performed better compared to your main competitors in the following areas?					
SD - Strongly Disagree, D – Disagree, N – Neutral, SA – Strongly Agree, A –Agree					
	SD	D	N	SA	A
16. Reduction of hazardous waste, emissions, etc.	☐	☐	☐	☐	☐
17. Consume less resources, such as energy, water, electricity, gas and petrol, etc.	☐	☐	☐	☐	☐
18. Compliance to environmental regulations	☐	☐	☐	☐	☐
19. Customer satisfaction in relation to product design and development	☐	☐	☐	☐	☐
20. Product design and innovation skill	☐	☐	☐	☐	☐
21. Quality of product and service	☐	☐	☐	☐	☐
22. Production cost	☐	☐	☐	☐	☐

Part four: company's profile

1. What is the company's main business activity?

- ☐ Food product and beverages
- ☐ Computer and electric products and components
- ☐ Electric products (eg: electric machines, ovens, fridges, washing machines and other electronic products).
- ☐ Chemical, rubber and plastic products
- ☐ Machinery and optical equipment
- ☐ Car and transport equipment
- ☐ Service
- ☐ Other, please specify. _____

2. Does your company have ISO14000?

- ☐ Yes, we are an ISO14000 certified company
- ☐ No, we are not an ISO14000 certified company

3. How many people does your company employ?

- ☐ under 200. ☐ between 200 and 399. ☐ between 400 and 599
- ☐ between 600 and 799. ☐ between 800 and 899. ☐ between 900 and 999
- ☐ over 1000

This is the end of the questionnaire.

Please check if there are any missing answers.

Thank you very much for your time and input.